PROGRAM 20

AIM:

To write a Prolog program to implement the Towers of Hanoi problem.

ALGORITHM:

1. Start the program.
2. Define the base case for one disk.
3. Move N-1 disks from source to auxiliary rod.
4. Move the largest disk from source to destination rod.
5. Move N-1 disks from auxiliary rod to destination rod.
6. Display the sequence of moves.
7. Stop the program.

PROGRAM:

```
hanoi(1, A, B, _) :-
    write('Move disk from '),
    write(A),
    write(' to '),
    write(B),
    nl.


hanoi(N, A, B, C) :-
    N > 1,
    N1 is N - 1,
    hanoi(N1, A, C, B),
    write('Move disk from '),
    write(A),
    write(' to '),
    write(B),
    nl,
    hanoi(N1, C, B, A).
```

OUTPUT:

```
25 ?- onsult(['c:/Users/Hema Neepa/OneDrive/Documents/Towers_of_Hunoi.pl']).
Correct to: "consult(['c:/Users/Hema Neepa/OneDrive/Documents/Towers_of_Hunoi.pl'])"?
Please answer 'y' or 'n'? yes
true.

26 ?- hanoi(3, left, right, middle).
Move disk from left to right
Move disk from left to middle
Move disk from right to middle
Move disk from left to right
Move disk from middle to left
Move disk from middle to right
Move disk from left to right
true
```

RESULT:

Thus, the Prolog program to implement the Towers of Hanoi was successfully executed.